

- 3 Two oscillating dippers are used to create progressive transverse water waves on the surface of a lake. The waves are shown from above in Figure 3.1.

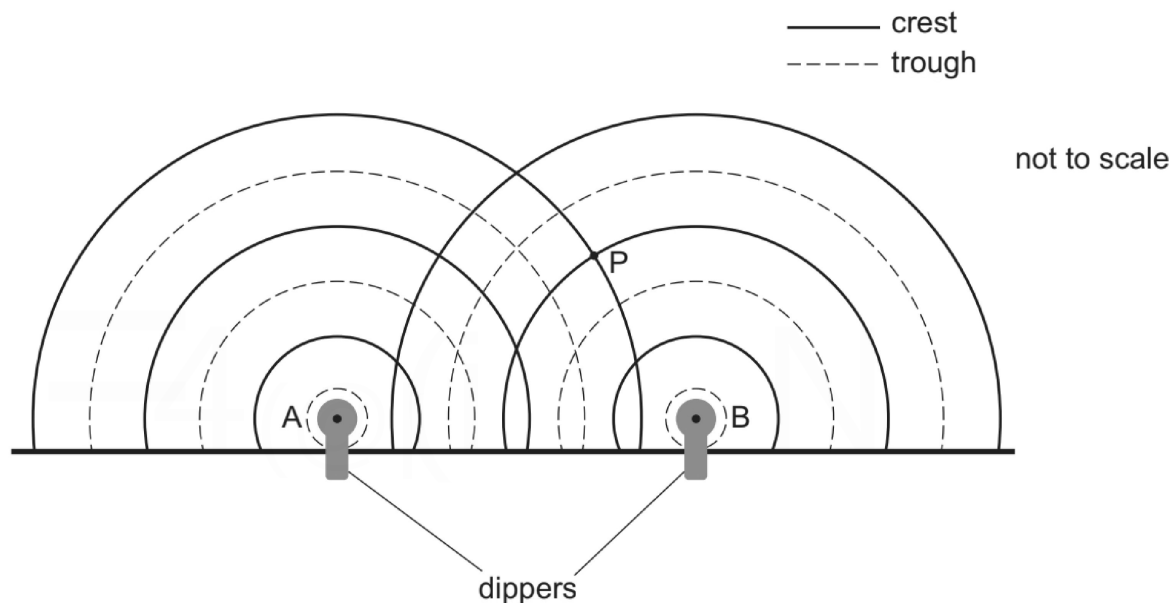


Figure 3.1

Wave crests are represented by a solid line. Wave troughs are represented by a dashed line. The waves superpose and an interference pattern is formed on the surface of the water.

The dippers oscillate in phase with each other and with a frequency of 3.0 Hz.

- (a) State what is meant by a progressive wave.

.....
 [1]

- (b) (i) Calculate the period of the waves.

period = s [1]

- (ii) Points A and B are at the centre of the dippers. Point P is shown on Figure 3.1. The distance from A to P is 3.9 m. The distance from B to P is 2.4 m.

Determine the speed of the waves.

speed = ms^{-1} [3]

- (c) A ball of mass 0.44 kg is placed at P, where it floats on the surface of the lake. The amplitude of the vertical oscillation of the ball is 15 cm.

- (i) Calculate the difference in the gravitational potential energy ΔE of the ball between its highest and lowest points.

$\Delta E = \dots\dots\dots$ J [3]

- (ii) The frequency of oscillation of the dippers is increased to 4.0 Hz without changing their amplitude of oscillation. The ball remains at the same position on the surface of the lake.

State and explain whether the amplitude of the oscillation of the ball increases, decreases or remains the same.

.....
.....
.....
..... [2]

[Total: 10]